

Math 142, §19
Exam 2

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	21	
2	24	
3	20	
4	20	
5	20	
Total	105	

Good Luck !

Problem 1. (21 points) Determine whether the **sequence** is convergent or divergent. If it is convergent, find its limit.

(a) $a_n = \frac{n^2 + n}{e^n}$

Let $f(x) = \frac{x^2 + x}{e^x}$, then $a_n = f(n)$

$$\begin{aligned} \lim_{x \rightarrow +\infty} f(x) &= \lim_{x \rightarrow +\infty} \frac{x^2 + x}{e^x} = \lim_{x \rightarrow +\infty} \frac{2x + 1}{e^x} \xrightarrow{\frac{\infty}{\infty}} \\ &= \lim_{x \rightarrow +\infty} \frac{2}{e^x} = 0 \end{aligned}$$

Thus $\{a_n\}$ is convergent and $\lim_{n \rightarrow +\infty} a_n = 0$

(b) $a_n = \frac{2 + n^3}{(n^2 - 6)(3n + 1)}$

$$\begin{aligned} \lim_{n \rightarrow +\infty} \frac{n^3 + 2}{(n^2 - 6)(3n + 1)} &= \lim_{n \rightarrow +\infty} \frac{n^3 + 2}{3n^3 + n^2 - 18n - 6} \cdot \frac{(1/n^3)}{(1/n^3)} \\ &= \lim_{n \rightarrow +\infty} \frac{1 + \frac{2}{n^3}}{3 + \frac{1}{n} - \frac{18}{n^2} - \frac{6}{n^3}} = \frac{1}{3} \end{aligned}$$

Thus $\{a_n\}$ is convergent and $\lim_{n \rightarrow +\infty} a_n = \frac{1}{3}$

(c) $a_n = (-1)^n \cos(\frac{1}{n})$

$$\lim_{n \rightarrow +\infty} \cos(\frac{1}{n}) = \cos(0) = 1$$

$$\text{Thus } \lim_{n \rightarrow +\infty} a_{2n} = \lim_{n \rightarrow +\infty} (-1)^{2n} \cos(\frac{1}{2n}) = 1$$

$$\lim_{n \rightarrow +\infty} a_{2n+1} = \lim_{n \rightarrow +\infty} (-1)^{2n+1} \cos(\frac{1}{2n+1}) = -1$$

$\Rightarrow \{a_n\}$ is divergent.

Problem 2. (24 points) Determine whether the series is convergent or divergent.

(a) $\sum_{n=1}^{+\infty} \frac{2^n}{3^n}$ (If convergent, please also find the sum for this series)

$$a_n = \frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n \rightarrow \text{a geometric series with } r = \frac{2}{3}$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{2^n}{3^n} \text{ is convergent since } |r| < 1.$$

$$\text{and } \sum_{n=1}^{+\infty} \frac{2^n}{3^n} = \frac{\frac{2}{3}}{1 - \frac{2}{3}} = 2$$

(b) $\sum_{n=2}^{+\infty} \frac{1}{n(\ln n)^3}$ (Hint: Use the Integration Test)

$$\text{let } f(x) = \frac{1}{x(\ln x)^3}, \text{ then } a_n = f(n)$$

$$\text{clearly } \frac{1}{n(\ln n)^3} > \frac{1}{(n+1)(\ln(n+1))^3} > 0 \text{ when } n \geq 2$$

$$\begin{aligned} \int_2^{+\infty} \frac{1}{x(\ln x)^3} dx &= \lim_{b \rightarrow +\infty} \int_2^b \frac{1}{x(\ln x)^3} dx = \lim_{b \rightarrow +\infty} \left[-\frac{1}{2(\ln x)^2} \right]_{x=2}^{x=b} \\ &= \lim_{b \rightarrow +\infty} -\frac{1}{2(\ln b)^2} + \frac{1}{2(\ln 2)^2} = \frac{1}{2(\ln 2)^2} < +\infty \end{aligned}$$

$$\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{n(\ln n)^3} \text{ is convergent.}$$

(c) $\sum_{n=1}^{+\infty} \frac{\sqrt{n^2+2}}{n^2+n+1}$ (Hint: Use the Limit Comparison Test)

$$\text{let } a_n = \frac{\sqrt{n^2+2}}{n^2+n+1}, \quad b_n = \frac{1}{n}.$$

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = \lim_{n \rightarrow +\infty} \frac{\sqrt{n^2+2}}{n^2+n+1} \cdot n = \lim_{n \rightarrow +\infty} \frac{n^2 \sqrt{1 + \frac{2}{n^2}}}{n^2(1 + \frac{1}{n} + \frac{1}{n^2})} = 1$$

$$\sum_{n=1}^{+\infty} \frac{1}{n} \text{ is divergent (by P-series with } P=1)$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{\sqrt{n^2+2}}{n^2+n+1} \text{ is divergent.}$$

Problem 3. (20 points) Determine whether the series is conditionally convergent, absolutely convergent or divergent.

(a) $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$

$$u_n = \frac{1}{\ln n}$$

clearly $u_n > 0$ when $n \geq 2$

i) $u_{n+1} = \frac{1}{\ln(n+1)} > \frac{1}{\ln n} > u_n$

ii) $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{\ln n} = 0$

$\Rightarrow \sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$ is convergent by Alternating series Test.

$|u_n| = \frac{1}{\ln n} > \frac{1}{n}$ when $n \geq 2 \Rightarrow \sum_{n=2}^{+\infty} |u_n| = \sum_{n=2}^{+\infty} \frac{1}{\ln n}$ is divergent.

thus, $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$ is conditionally convergent.
 Since $\sum_{n=2}^{+\infty} \frac{1}{n}$ is divergent.

(b) $\sum_{n=1}^{+\infty} \frac{(-1)^n e^{\frac{1}{n}}}{n^2}$

$$u_n = \frac{e^{\frac{1}{n}}}{n^2}, \quad b_n = \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{b_n} = \lim_{n \rightarrow \infty} \frac{e^{\frac{1}{n}}}{n^2} \cdot n^2 = \lim_{n \rightarrow \infty} e^{\frac{1}{n}} = 1$$

$\sum_{n=1}^{+\infty} \frac{1}{n^2}$ is convergent $\Rightarrow \sum_{n=1}^{+\infty} \frac{e^{\frac{1}{n}}}{n^2}$ is convergent by

Limit Comparison Test

$\Rightarrow \sum_{n=1}^{+\infty} (-1)^n \frac{e^{\frac{1}{n}}}{n^2}$ is absolutely convergent.

Problem 4. (20 points) Find the radius of convergence and interval of convergence of the power series. (Hint: Use the Ratio Test and then check the two end points)

(a) $\sum_{n=1}^{+\infty} \frac{(-1)^n x^n}{n 2^n}$

$$\begin{aligned} \lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow +\infty} \left| \frac{(-1)^{n+1} x^{n+1}}{(n+1) 2^{n+1}} \cdot \frac{n 2^n}{(-1)^n x^n} \right| \\ &= \lim_{n \rightarrow +\infty} \left| \frac{x}{2} \cdot \frac{n}{n+1} \right| = \left| \frac{x}{2} \right| < 1 \Rightarrow |x| < \underline{2} \Rightarrow R = 2 \text{ or } x \in (-2, 2). \end{aligned}$$

Check two end points $x = -2, 2$.

i) $x = -2$, $a_n = \frac{(-1)^n (-2)^n}{n 2^n} = \frac{1}{n}$, $\sum_{n=1}^{+\infty} a_n = \sum_{n=1}^{+\infty} \frac{1}{n}$ diverges.

ii) $x = 2$, $a_n = \frac{(-1)^n 2^n}{n 2^n} = (-1)^n \frac{1}{n}$, $\sum_{n=1}^{+\infty} a_n = \sum_{n=1}^{+\infty} (-1)^n \frac{1}{n}$ converges.

Thus the interval of convergence is $(-2, 2]$

(b) $\sum_{n=0}^{+\infty} (n+1)(x-1)^n$

$$\begin{aligned} \lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow +\infty} \left| \frac{(n+2)(x-1)^{n+1}}{(n+1)(x-1)^n} \right| = \lim_{n \rightarrow +\infty} \left| \frac{n+2}{n+1} \cdot (x-1) \right| \\ &= |x-1| < \underline{1} \Rightarrow R = 1 \end{aligned}$$

$$|x-1| < 1 \Rightarrow -1 < x-1 < 1 \Rightarrow 0 < x < 2.$$

Two end points $x = 0, 2$

i) $x = 0$, $a_n = (n+1)(-1)^n$, $\sum_{n=0}^{+\infty} (n+1)(-1)^n$ diverges.

ii) $x = 2$, $a_n = (n+1)1^n = n+1$, $\sum_{n=0}^{+\infty} n+1$ diverges.

Thus the interval of convergence is $(0, 2)$.

Problem 5. (20 Points) Find (**and simplify if possible**) the Maclaurin series (the power series centered at 0) and its radius of convergence for the following functions.

(a) $f(x) = \frac{1}{2}(e^x + e^{-x})$. (Hint: Use $e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!}$ for $|x| < +\infty$)

$$\begin{aligned} f(x) &= \frac{1}{2}(e^x + e^{-x}) \\ &= \frac{1}{2} \left(\sum_{n=0}^{+\infty} \frac{x^n}{n!} + \sum_{n=0}^{+\infty} \frac{(-x)^n}{n!} \right) \\ &= \frac{1}{2} \sum_{n=0}^{+\infty} \frac{x^n + (-1)^n x^n}{n!} = \frac{1}{2} \sum_{n=0}^{+\infty} \frac{1 + (-1)^n}{n!} x^n \\ &= \frac{1}{2} \sum_{n=0}^{+\infty} \frac{2}{(2n)!} x^{2n} \\ &= \sum_{n=0}^{+\infty} \frac{1}{(2n)!} x^{2n}, \quad \text{for any } |x| < +\infty \Rightarrow R = +\infty \end{aligned}$$

(b) $f(x) = \frac{x^2}{3-x}$ (Hint: Use $\frac{1}{1-x} = \sum_{n=0}^{+\infty} x^n$ for $|x| < 1$)

$$\begin{aligned} f(x) &= \frac{x^2}{3-x} = \frac{x^2}{3(1-\frac{x}{3})} = \frac{x^2}{3} \cdot \frac{1}{1-\frac{x}{3}} = \frac{x^2}{3} \sum_{n=0}^{+\infty} \left(\frac{x}{3}\right)^n \\ &= \sum_{n=0}^{+\infty} \frac{x^2}{3} \cdot \frac{x^n}{3^n} = \sum_{n=0}^{+\infty} \frac{x^{n+2}}{3^{n+1}} \\ &= \sum_{n=2}^{+\infty} \frac{x^n}{3^{n-1}} \quad \text{for } \left|\frac{x}{3}\right| < 1 \Rightarrow |x| < 3 \Rightarrow R=3 \end{aligned}$$